AppUser

Column	Constraint / Reference
UserID	PK
Name	
Email	
Phone	
UserStatus	
MembershipTier	
Region	
City	
Street	

Librarian

Column	Constraint / Reference
LibID	PK
Name	
Email	

Category

Column	Constraint / Reference
CID	PK
CName	
ParentCID	FK → Category.CID; nullable

PromoCode

Column	Constraint / Reference
PromoID	PK
Code	UNIQUE
Percentage	
StartDate	
EndDate	

PromoCategory

Column	Constraint / Reference
PromoID	PK, FK → PromoCode.PromoID
CID	PK, FK → Category.CID

Tool

Column	Constraint / Reference
ToolID	PK
TName	
Condition	
Status	
PPerHour	
PPerDay	
PPerWeek	
UsageHours	
InspectionThreshold	
CleaningBufferHours	
OwnerID	FK → AppUser.UserID
CID	FK → Category.CID

ToolDocument

Column	Constraint / Reference
ToolID	PK, FK → Tool.ToolID
DocNo	PK
Type	
FileURL	
WarrantyExpiry	Nullable

Reservation

Column	Constraint / Reference
TransID	PK
BookingDate	
Status	
BorrowerID	FK → AppUser.UserID

ReservedItem

Column	Constraint / Reference
TransID	PK, FK → Reservation.TransID
ToolID	PK, FK → Tool.ToolID
Price	
RentDate	
ReturnDate	
ActualReturnDate	Nullable
HandoverDate	Nullable
QrToken	UNIQUE
PromoID	FK → PromoCode.PromoID; nullable

Deposit

Column	Constraint / Reference
DepID	PK
HeldAmount	
Status	
TransID	UNIQUE, FK → Reservation.TransID

Review

Column	Constraint / Reference
RevID	PK
RevStars	
RevDate	
Comment	
ReviewerID	FK → AppUser.UserID
RevieweeID	FK → AppUser.UserID
TransID	FK → Reservation.TransID
(TransID, ReviewerID)	UNIQUE

Message

Column	Constraint / Reference
MsgID	PK
MsgContent	
TimeSent	
SenderID	FK → AppUser.UserID
ReceiverID	FK → AppUser.UserID

Dispute

Column	Constraint / Reference
DisID	PK
Type	
Description	
Status	
FiledDate	
InsuranceStatus	
RefundedAmount	Nullable
FilerID	FK → AppUser.UserID
TransID	FK → Reservation.TransID
LibID	FK → Librarian.LibID; nullable

DisputePhoto

Column	Constraint / Reference
DisID	PK, FK → Dispute.DisID
Photo	PK

Penalty

Column	Constraint / Reference
PenID	PK
Level	
FineAmount	Nullable
Reason	
StartDate	
EndDate	Nullable
UserID	FK → AppUser.UserID
TransID	Composite FK → ReservedItem; nullable
ToolID	Composite FK → ReservedItem; nullable
LibID	FK → Librarian.LibID; nullable

Audit

Column	Constraint / Reference
LibID	PK, FK → Librarian.LibID
ToolID	PK, FK → Tool.ToolID
ReqDate	PK
Status	
Photo	Nullable

MaintenanceRecord

Column	Constraint / Reference
ToolID	PK, FK → Tool.ToolID
RecNo	PK
Type	
Date	
Description	
Result	
Status	
ExternalShop	Nullable
LibID	FK → Librarian.LibID; nullable